Exercise Sheet #7: Phasor Representation

Discussion: Dec 1, 2025

Exercise 1: Sinusoidal Voltage

Given the sinusoidal voltage $v(t) = 50\cos(30t + 10^\circ)\,\text{V}$, find:

a) the amplitude V_m

b) the period T

c) the frequency f

d) $v(t)$ at $t = 10\,\text{ms}$.

Solution

a) Amplitude:

$$V_m = 50\,\text{V}$$

b) Period:

$$T = \frac{2\pi}{\omega} = \frac{2\pi}{30} \approx 209.4\,\text{ms}$$

c) Frequency:

$$f = \frac{1}{T} = \frac{30}{2\pi} \approx 4.775\,\text{Hz}$$

d) Voltage at $t = 10\,\text{ms}$:

$$v(0.01) = 50\cos(30 \cdot 0.01 + 10^\circ)$$

Convert 10° to radians:

$$10^\circ = \frac{10\pi}{180} = 0.17453\,\text{rad}$$

Compute the angle:

$$30 \cdot 0.01 + 0.17453 = 0.47453\,\text{rad}$$

Thus:

$$v(0.01) = 50\cos(0.47453) \approx 44.48\,\text{V}$$

Exercise 2: Phase Shift

Given $v_1 = 45\sin(\omega t + 30^\circ)\,\text{V}$ and $v2 = 50\cos(\omega t - 30^\circ)\,\text{V}$, determine the phase angle between the two sinusoids and which one lags the other.

Solution

Convert v_2 to sine form:

$$v_2 = 50\cos(\omega t - 30^\circ) = 50\sin(\omega t - 30^\circ + 90^\circ) = 50\sin(\omega t + 60^\circ)$$

Thus the phase angles are

$$\phi_1 = 30^\circ, \qquad \phi_2 = 60^\circ$$

Phase difference:

$$\Delta\phi = \phi_2 - \phi_1 = 60^\circ - 30^\circ = 30^\circ$$

Since $\phi_1 < \phi_2$,

$$v_1 \text{ lags } v_2 \text{ by } 30^\circ.$$

Exercise 3: ...

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Solution

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Exercise 4: ...

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Solution

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Exercise 5: ...

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Solution

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Exercise 6: ...

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Solution

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Exercise 7: ...

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Exercise 8: ...

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Solution

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